# Univariate Stationary Time Series

**References:**

Tsay (2010), Ch. 2; Enders (2014), Ch. 2; Hamilton (1994), Ch. 3; Pesaran (2015), Chs. 12, 14

## 1 Stationarity

**Definition 1.1. (strict stationarity)** A time series $\{y_t\}$ is said to be *strictly stationary* if the joint distribution of $(y_{t_1}, y_{t_2}, \ldots, y_{t_k})$ is identical to that of $(y_{t_1+t}, y_{t_2+t}, \ldots, y_{t_k+t})$ for all $t$, where $k$ is an arbitrary positive integer and $(t_1, t_2, \ldots, t_k)$ is a collection of $k$ positive integers.

- Strict stationarity requires that the joint distribution of $(y_{t_1}, y_{t_2}, \ldots, y_{t_k})$ is invariant under time shift.
- A process under strict stationarity is in "stochastic equilibrium", which is a counterpart of the static equilibrium concept in deterministic processes.
- This is a very strong assumption and hard to verify empirically.

**Definition 1.2. (weak stationarity)** A time series $\{y_t\}$ is said to be *weakly stationary* if

(a) it has a constant mean, namely, $E(y_t) = \mu$, which is a constant, and

(b) its covariance function, $\gamma(t_1, t_2)$, defined by

$$\gamma(t_1, t_2) = Cov(y_{t_1}, y_{t_2}) = E(y_{t_1} y_{t_2}) - E(y_{t_1}) E(y_{t_2}),$$

depends only on $|t_1 - t_2|$, namely, $\gamma(t_1, t_2) = \gamma(|t_1 - t_2|)$.

- Conditions (b) states that the covariance between any two observations depends only on the length of time separating the observations.
- Condition (b) implies that $\{y_t\}$ has a constant variance.

- In the definition of weak stationarity, we implicitly assumed that the first two moments of $y_t$ are finite. If $y_t$ is strictly stationary and its first two moments are finite, then $y_t$ is also weakly stationary. The converse is not true in general with the exception of normally distributed series.

- A weakly stationary process is also referred to as *covariance stationary*, *second-order stationary*, or *wide-sense stationary* (or, simply, stationary).

- The weak stationarity assumption can be checked empirically. A visual examination of a weakly stationary time series will find the data fluctuating around a fixed level with constant variation.

**Definition 1.3.** A time series $\{y_t\}$ is said to be *trend stationary* if $y_t = f(t) + x_t$, where $f(t)$ is a deterministic function of $t$, and $x_t$ is covariance stationary.

- $f(t)$ is also known as the "perfectly predictable component" of $y_t$.

- Examples of $f(t)$ include linear trend, $f(t) = a_0 + a_1 t$, and seasonal dummies, $f(t) = d_t = 1$ for some $t$.

**Definition 1.4.** A time series $\{\varepsilon_t\}$ is said to be a *white noise* process if it has zero mean, a constant variance $\sigma^2$, and $\varepsilon_t$ are uncorrelated over time, namely, $E(\varepsilon_t \varepsilon_{t'}) = 0$, for $t \neq t'$. For simplicity, we often denote it as $\varepsilon_t \sim WN(0, \sigma^2)$.

- Intuition: If $\varepsilon_t$ is unusually high, $\varepsilon_{t+1}$ has no tendency to be unusually high or low.

- Some books define white noise with a slightly stronger condition, requiring $\varepsilon_t$ to be independent, instead of uncorrelated, over time.

- If $\varepsilon_t \sim IIDN(0, \sigma^2)$, it is referred to as a *Gaussian white noise* process.

## 2 Autocorrelation Function

- Recall that the correlation coefficient between two random variables $X$ and $Y$ is defined as

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sqrt{E(X-\mu_X)^2\,E(Y-\mu_Y)^2}}.$$

  - $\rho_{X,Y}$ measures the strength of linear dependence between $X$ and $Y$
  - $-1 \leq \rho_{X,Y} \leq 1$
  - Independent of $X$ and $Y \Rightarrow \rho_{X,Y} = 0$
  - If both $X$ and $Y$ are normal random variables, then independent of $X$ and $Y \Leftrightarrow \rho_{X,Y} = 0$

- Given sample $\{x_t, y_t\}_{t=1}^T$, the sample covariance function is

$$\widehat{Cov}(X,Y) = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{T-1}, \tag{1}$$

  where $\bar{x} = T^{-1}\sum_{t=1}^T x_t$ and $\bar{y} = T^{-1}\sum_{t=1}^T y_t$. The sample correlation function is given by

$$\hat{\rho}_{X,Y} = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_{t=1}^T (x_t - \bar{x})^2}\sqrt{\sum_{t=1}^T (y_t - \bar{y})^2}}.$$

  It can be shown that $\widehat{Cov}(X,Y) \to_p Cov(X,Y)$ and $\hat{\rho}_{X,Y} \to_p \rho_{X,Y}$, as $T \to \infty$.

The concept of correlation can be applied to autocorrelation.

**Definition 2.1.** Consider a weakly stationary series $\{y_t\}$. The *autocovariance function* of $\{y_t\}$ at lag $h$ (or lag-$h$ autocovariance) is

$$\gamma_h = Cov(y_{t+h}, y_t), \text{ for } h = 0, \pm 1, \pm 2, \ldots.$$

The *autocorrelation function (ACF)* of $\{y_t\}$ at lag $h$ is defined as

$$\rho_h = \frac{Cov(y_{t+h}, y_t)}{\sqrt{Var(y_{t+h})\,Var(y_t)}} = \frac{Cov(y_{t+h}, y_t)}{Var(y_t)} = \frac{\gamma_h}{\gamma_0}.$$

- $\gamma_0 = Var(y_t) > 0$, $\gamma_h = \gamma_{-h}$, $|\gamma_h| \leq \gamma_0$, for all $h$.
- $\rho_0 = 1$, $\rho_h = \rho_{-h}$, $-1 \leq \rho_h \leq 1$, for all $h$.
- A weakly stationary series $\{y_t\}$ is *serially uncorrelated* if and only if $\rho_h = 0$ for all $h > 0$.

**Definition 2.2.** Consider a weakly stationary series $\{y_t\}_{t=1}^T$. The *sample autocovariance function* of $\{y_t\}$ at lag $h$ (or lag-$h$ autocovariance) is

$$\hat{\gamma}_h = \frac{\sum_{t=h+1}^T (y_t - \bar{y})(y_{t-h} - \bar{y})}{T}, \ 0 \leq h \leq T-1, \tag{2}$$

where $\bar{y} = T^{-1}\sum_{t=1}^T y_t$. The *sample autocorrelation function (sample ACF)* is

$$\hat{\rho}_h = \frac{\hat{\gamma}_h}{\hat{\gamma}_0} = \frac{\sum_{t=h+1}^T (y_t - \bar{y})(y_{t-h} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}, \ 0 \leq h \leq T-1.$$

- It can be proved that $\hat{\gamma}_h \to_p \gamma_h$, for a fixed $h$ and as $T \to \infty$.
- Comparing (2) with (1), we see that $\hat{\gamma}_h$ is approximately the sample covariance of the $T-h$ pairs of observations $(y_1, y_{h+1})$, $(y_2, y_{h+2})$, ..., $(y_{T-h}, y_T)$, but note that $\hat{\gamma}_h$ uses the *overall* mean, $\bar{y}$, and denominator $T$, instead of $T-h$.
- Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, and recall that $\gamma_h = Cov(y_t, y_{t-h})$. Then

$$Var(\mathbf{y}) = \begin{pmatrix} Var(y_1) & Cov(y_1,y_2) & \ldots & Cov(y_1,y_{T-1}) & Cov(y_1,y_T) \\ Cov(y_2,y_1) & Var(y_2) & \ldots & Cov(y_2,y_{T-1}) & Cov(y_2,y_T) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ Cov(y_{T-1},y_1) & Cov(y_{T-1},y_2) & \ldots & Var(y_{T-1}) & Cov(y_{T-1},y_T) \\ Cov(y_T,y_1) & Cov(y_T,y_2) & \ldots & Cov(y_T,y_{T-1}) & Var(y_T) \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{T-2} & \gamma_{T-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{T-3} & \gamma_{T-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_{T-2} & \gamma_{T-3} & \cdots & \gamma_0 & \gamma_1 \\ \gamma_{T-1} & \gamma_{T-2} & \cdots & \gamma_1 & \gamma_0 \end{pmatrix}.$$

- Using denominator $T$ instead of $T-h$ in (2) ensures the positive definiteness of the sample covariance matrix.

- If $\{y_t\}$ is an IID series with $E\left(y_t^2\right) < \infty$, then

$$\hat{\rho}_h \sim N\left(0, \frac{1}{T}\right), \text{ or equivalently, } \sqrt{T}\hat{\rho}_h \sim N\left(0,1\right),$$

  as $T \to \infty$, for any fixed positive integer $h$.

- More generally, if $\{y_t\}$ is a weakly stationary time series satisfying $y_t = \mu + \sum_{i=0}^{q} \alpha_i \varepsilon_{t-i}$, where $\varepsilon_i \sim IID\left(0, \sigma^2\right)$ and $\alpha_0 = 1$, then

$$\hat{\rho}_h \sim N\left(0, \frac{1 + 2\sum_{i=1}^{q} \rho_i^2}{T}\right), \text{ for } h > q. \tag{3}$$

- In finite samples, $\hat{\rho}_h$ is a biased estimator of $\rho_h$, and the bias is in the order of $1/T$.

- **Testing individual ACF**: For a given positive integer $h$, the test statistic for testing $H_0 : \rho_h = 0$ against $H_1 : \rho_h \neq 0$ is

$$t \text{ ratio} = \frac{\hat{\rho}_h}{\sqrt{\left(1 + 2\sum_{i=1}^{h-1} \hat{\rho}_i^2\right)/T}}.$$

  For simplicity, many software packages use $1/T$ as the asymptotic variance of $\rho_h$ for all $h \neq 0$, assuming that the underlying time series is an IID sequence.

- Instead of testing randomness at each individual lag, we may want to evaluate the overall randomness based on a number of lags.

- **Portmanteau Test**: $H_0$: $\rho_1 = \rho_2 = \ldots = \rho_m = 0$ against $H_1$: $\rho_i \neq 0$, for some $i \in \{1, 2, \ldots, m\}$. If $\{y_t\}$ is a finite-variance IID series, then

  – Box and Pierce (1970) test statistic (of order $m$):

$$Q^*\left(m\right) = T\sum_{h=1}^{m} \hat{\rho}_h^2 \sim \chi^2\left(m\right)$$

– Ljung and Box (1978) test statistic (of order $m$), which is a small-sample modification of $Q^*(m)$, is given by:

$$Q(m) = T(T+2)\sum_{h=1}^{m}\frac{\hat{\rho}_h^2}{T-h} \sim \chi^2(m)$$

– Decision rule: Reject $H_0$ if $Q(m) > \chi^2_\alpha(m)$, where $\chi^2_\alpha(m)$ denotes the $100(1-\alpha)$ th percentile of a chi-squared distribution with $m$ degrees of freedom, or $p$-value $\leq \alpha$, where $\alpha$ is a given significance level.

# 3 Introduction to ARMA Models

- Let $\{\varepsilon_t\}$ be a white noise process with zero mean and constant variance $\sigma^2$. We will explore a class of models formed by linear combinations of white noise.
- Autoregressive (AR) processes
  - The *pth-order autoregressive process*, denoted by $AR(p)$:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t, \tag{4}$$

    where $\phi_0$, $\phi_1$,...,$\phi_p$ are constants.
  - For example, when $p = 1$, we have the *first-order autoregressive process*, denoted by $AR(1)$:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t.$$

- Moving average (MA) processes
  - The *qth-order moving average process*, denoted by $MA(q)$:

$$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_q\varepsilon_{t-q}, \tag{5}$$

    where $\mu, \theta_1, \theta_2, \ldots, \theta_q$ are constants.

  - For example, when $q = 1$, we have the *first-order moving average process*, denoted by $MA(1)$:

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1}.$$

- Mixed autoregressive moving average processes
  - $ARMA(p, q)$:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}. \quad (6)$$

- It will be easy to represent and manipulate ARMA models with the polynomial *lag operators*. Denote the first-order lag operator by $L$, which moves the time index back one unit:

$$L y_t = y_{t-1}$$

  By repeated application of the operator, we have

$$L^2 y_t = L L y_t = L y_{t-1} = y_{t-2},$$
$$L^j y_t = y_{t-j}, \quad L^{-j} y_t = y_{t+j}, \quad L^0 = 1.$$

- Using the lag operator notation, the $AR(p)$ model given by (4) can be written as

$$\left(1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p\right) y_t = \phi_0 + \varepsilon_t,$$

  or

$$\phi(L)\, y_t = \phi_0 + \varepsilon_t,$$

  where $\phi(L)$ is a *lag polynomial* defined by

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p = 1 - \sum_{i=1}^{p} \phi_i L^i.$$

- Similarly, the $MA(q)$ model given by (5) can be written as

$$y_t = \mu + \left(1 + \theta_1 L + \theta_2 L^2 + \ldots + \theta_q L^q\right) \varepsilon_t,$$

or simply

$$y_t = \mu + \theta(L)\,\varepsilon_t,$$

where $\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \ldots + \theta_q L^q = \sum_{i=0}^{q} \theta_i L^i$, with $\theta_0 \equiv 1$.

- The $ARMA(p, q)$ process given by (6)

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}$$

can be written as

$$\phi(L)\, y_t = c + \theta(L)\, \varepsilon_t.$$

# 4 Moving Average Processes

## 4.1 Finite-Order MA Models

- Consider the $MA(1)$ model:

$$y_t = \varepsilon_t + \theta \varepsilon_{t-1},$$

where $\{\varepsilon_t\}$ is a white noise process with zero mean and constant variance $\sigma^2$, or $\varepsilon_t \sim WN(0, \sigma^2)$. To simplify the exposition, we have assumed that $\mu = 0$. The discussion can be readily extended to series with non-zero means and other deterministic trends.

  - *Exercise*: Show that its mean is zero and its autocorrelation function is given by

$$\rho_0 = 1, \quad \rho_1 = \frac{\theta}{1 + \theta^2}, \quad \text{and} \quad \rho_h = 0, \text{ for } h > 1.$$

*Hint*: Recall that $E(\varepsilon_t \varepsilon_s) = 0$, for $t \neq s$.

- Consider the $MA(2)$ model:

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

  - *Exercise*: Show that its autocorrelation function is

$$\rho_0 = 1, \quad \rho_1 = \frac{\theta_1 + \theta_1 \theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_2 = \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \text{and} \quad \rho_h = 0, \text{ for } h > 2.$$

- Do you see the pattern? For an $MA(q)$ process, $y_t = \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}$, only the first $q$ autocovariances/autocorrelations are non-zero.

$$\gamma_h = \begin{cases} \sigma^2 \sum_{i=0}^{q-|h|} \theta_i \theta_{i+|h|} & \text{if } 0 \leq |h| \leq q \\ 0 & \text{if } |h| > q \end{cases}$$

- $MA(q)$ is a "finite memory" model—it is only linearly related to its first $q$-lagged values.

## 4.2 Infinite-Order MA Models

- Can we let $q \to \infty$? The *infinite-order moving average process*, denoted by $MA(\infty)$, is defined as:

$$y_t = \lim_{q \to \infty} \sum_{i=0}^{q} \theta_i \varepsilon_{t-i} = \sum_{i=0}^{\infty} \theta_i \varepsilon_{t-i} = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots, \tag{7}$$

  For the infinite series $\sum_{i=0}^{q} \theta_i \varepsilon_{t-i}$ to converge to a well-defined limit as $q \to \infty$, certain restrictions must be imposed on the coefficients (weights) $\{\theta_i\}$ .

- (7) is a well-defined weakly stationary process[1] if

$$\sum_{i=0}^{\infty} \theta_i^2 < \infty. \tag{8}$$

  It is convenient to work with a slightly stronger condition than (8):

$$\sum_{i=0}^{\infty} |\theta_i| < \infty. \tag{9}$$

- A sequence of numbers $\{\theta_i\}_{i=0}^{\infty}$ satisfying (8) is said to be *square summable*, whereas a sequence satisfying (9) is said to be *absolutely summable*.

- Absolute summability implies square summability, but the reverse does not hold.

- Note that $Var(y_t) = \sigma^2 \sum_{i=0}^{\infty} \theta_i^2 < \infty$. What are the mean, autocovariance, autocorrelation of the $MA(\infty)$ process in (7)?

- An $MA(\infty)$ process with absolutely summable coefficients will have absolutely summable autocovariances: $\sum_{h=0}^{\infty} |\gamma_h| < \infty$.

[1] The $MA(\infty)$ exists in the mean squared sense $\lim_{q \to \infty} E\left(\left|y_t - \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}\right|^2\right) = 0$.

*Exercise*:

1. Consider the $MA(1)$ model: $y_t = 5 + \varepsilon_t + .7\varepsilon_{t-1}$, where $\varepsilon_t \sim IIDN(0,1)$. Find the autocorrelation function of $y_t$.

2. Consider the $MA(2)$ model: $y_t = 3 + \varepsilon_t + .6\varepsilon_{t-1} + .4\varepsilon_{t-2}$, where $\varepsilon_t \sim IIDN(0,1)$. Find the autocorrelation function of $y_t$.

# 5 Autoregressive Processes

## 5.1 AR(1) Model

- Consider the $AR(1)$ model:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \tag{10}$$

  where $\varepsilon_t \sim WN(0, \sigma^2)$. Let us find the necessary and sufficient condition for weak stationarity of the $AR(1)$ model.

- Assuming that $\{y_t\}$ is weakly stationary, then

$$E(y_t) = \phi_0 + \phi_1 E(y_{t-1}) + E(\varepsilon_t).$$

  Since $E(y_t) = E(y_{t-1}) = \mu$ under stationarity, we have

$$\mu = \phi_0 + \phi_1 \mu, \quad \text{or} \quad E(y_t) = \mu = \frac{\phi_0}{1 - \phi_1}.$$